



Sri Hari Vishnubhatla

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Summary

Fluid mechanics researcher with mechanical engineering background and cross-domain experience in embedded electronics and computer science. My doctoral research focused on uncovering the physics of drag modulation in particle-laden turbulent pipe flows experimentally. I utilized optical diagnostic techniques (PIV, PTV, PLIF) to explore mean flow feature modifications. Developed robust methodologies for image segmentation in multiphase flows, PIV-based measurement of mean wall shear stress, iterative method for attenuation compensation for accurate concentration measurements in PLIF, and a low cost programmable timing unit (PTU) to overcome experimental limitations and diagnostic bottlenecks. I am a collaborative problem-solver experienced in troubleshooting complex experimental rigs, designing custom synchronization hardware, and aiding in development of streamlined data analysis pipelines for peers. Dedicated to building open-source experimental tools and advancing the fidelity and accessibility of fluid mechanics research.

Areas of Interest

My areas of research interests broadly include:

Experimental measurements of turbulent flows · Internal Turbulent Flows (Pipe, Channel etc.) · Flow diagnostics and control · Flow visualization · Turbulent boundary layer measurements and control · Pollutant Dispersion · Optical Flow Measurement (PIV/PTV/PLIF/QSV) · Vehicle Aerodynamics and Design Optimization · Passive and Active Drag Control.

Education

Indian Institute of Technology (IIT) Delhi

New Delhi, India

Ph.D. in Experimental Fluid Mechanics, Dept. of Applied Mechanics

2020 – Present

– **Advisor:** Prof. Murali R. Cholemari

– **Thesis Title:** Drag Reduction in Internal Turbulent Flows using Rigid Spherical Particles as Shear Free Surfaces

– **Coursework:** Applied Fluid Dynamics, Advanced Fluid Mechanics, Physics of Turbulent Flows, Turbulence and its Modelling, Computational Fluid Dynamics (CFD), Experimental Methods, Engineering Mathematics & Computation.

Sant Longowal Institute of Engineering and Technology (SLIET)

Longowal, Punjab, India

Bachelor of Engineering (B.E.), Mechanical Engineering (spl. Manufacturing Engineering)

2016 – 2020

Graduation grade: 9.15 on a scale of 10 · First Division with Distinction

Kendriya Vidyalaya

University of Hyderabad, Telangana, India

Senior Secondary High School (Class XII)

2015 – 2016

Graduation grade: 93% · School Topper

Teaching & Mentorship

Teaching Assistant

IIT Delhi

Department of Applied Mechanics

2021 – Present

– Facilitated laboratory sessions, graded coursework, and held office hours for core and advanced departmental courses including: *Fluid Mechanics, Advanced Fluid Mechanics, Physics of Turbulent Flows, Modeling of Solids & Fluids, and Experimental Methods.*

– Served as lead TA for *Advanced Experimental Methods*, directly instructing B.Tech students in the hands-on setup, calibration, and operation of complex optical flow measurement techniques including PIV and PLIF.

– Developed and deployed course websites for Fluid Mechanics and Physics of Turbulent Flows courses for easier and organized dissemination of course materials.

Experience and Projects

Research Experience

Passive Drag Reduction In Internal Turbulent Flows

IIT Delhi

PhD thesis under Prof. Murali R. Cholemani

Sep 2020 – Present

This was my PhD Thesis at IIT Delhi titled ‘Drag Reduction in Internal Turbulent Flows using Rigid Spherical Particles as Shear Free Surfaces’. The efficacy of rigid particles in reducing turbulent drag in pipe flows was investigated. Three experimental setups including a vertical pipe flow facility, horizontal pipe flow facility and a turbulent convection setup were developed and utilized to aid in the investigation. Particle Image Velocimetry was the primary diagnostic tool for this. The core hypothesis that significant drag reduction can be brought about by rigid particles when they behave as “Shear Free Surfaces” is experimentally tested and validated.

Flux Enhancement in Turbulent Convection using SFSES

IIT Delhi

assisted Adhitya Krishnan (MSR) with Prof. Murali R. Cholemani

Aug 2021 – May 2023

The enhancement in exchange flux in a turbulent convection in a long tube at high Rayleigh numbers due to the introduction of Shear Free Surfaces (SFSES) was experimentally studied. Simultaneous PIV+PLIF diagnostics revealed that the reduction in surface normal turbulent transport was the root cause of such an effect.

Dispersion of passive scalar in a vehicle wake

IIT Delhi

assisted Manish K. Mathur (PhD) with Prof. Murali R. Cholemani

Aug 2023 – October 2025

Experiments were conducted in a wind tunnel with rolling road to evaluate the dispersion of passive scalar in the wake of a moving vehicle. A novel technique called Quantitative Smoke Visualization was developed for this investigation. My specific contribution was in the development of the said technique, processing algorithm and scripts for iterative correction for attenuation compensation of the scattered concentration signal intensity, and the deployment of a custom Programmable Timer Unit (PTU) for concentration and velocity field measurements.

Tools Developed as part of Research

Near Wall Projection Method to Measure Wall Shear Stress using PIV

IIT Delhi

To overcome instrumentation limitations in determining local mean wall shear stress

2025

Developed and implemented a technique leveraging known near wall physics to quantify the mean wall shear stress using reliable off wall PIV measurements. Utilizing the developed functional form of the non-dimensional mean velocity gradient, the measurements are projected onto the wall to circumvent the lacunae of reliable measurements in the immediate vicinity of the wall.

Intensity Based Image Segmentation (IBIS)

IIT Delhi

For segmenting tracer particles from dispersed phase in a particle laden flow

2023

Developed and implemented an Intensity Based Image Segmentation (IBIS) algorithm to accurately identify and isolate dispersed particle phase from tracer particles in complex, particle-laden flow images. The algorithm utilizes a set of sequential top hat sliding filters and Gaussian blur to successfully eliminate the halos around particles and isolate the secondary scatter due to aplanar particles.

Iterative Technique for Attenuation Compensation

IIT Delhi

for accurate concentration measurements in PLIF without a reference cell

2024

Developed an iterative technique for attenuation compensation in Planar Laser Induced Fluorescence (PLIF) to accurately quantify scalar concentration fields without the need for a reference cell. By applying the iterative process to the ensemble mean of the fluorescence field, the method achieves extremely rapid, monotonic convergence. This approach minimizes systematic underprediction of concentration due to laser intensity attenuation offering a robust solution for high-fidelity measurements in complex, optically thick flows. This serves as the foundational tool for accurate concentration measurements in gaseous flow using the **Quantitative Smoke Visualization** (QSV) technique.

Projects

OpenPTU: A Custom Programmable Timing Unit

IIT Delhi

To interface hardware beyond proprietary ecosystems

2023

Designed and implemented an AtMega328p-based Programmable Timer Unit (PTU) to achieve sub-microsecond-level precise hardware synchronization. Enabled TR-PIV using off-the-shelf high-speed camera and dual cavity pulsed laser providing a cost-effective solution for experimental diagnostics.

Design, Modeling and Analysis of an Aircraft Wing

SLIET Longowal

Bachelor of Engineering Project with Dr. Amrik Singh

2019-2020

Engineered an optimized Low Altitude Long Endurance wing using two-way coupled fluid-structure interaction to ensure structural stability and aerodynamic efficiency in turbulent, low-altitude environments.

Automation in Agriculture

SLIET Longowal

with Team Endeavour under Dr. Amrik Singh

2018-2020

Designed an automated sensor-based system to monitor and maintain optimal environmental conditions (CO₂, humidity, temperature, soil moisture) for maximized seasonal crop yield. Project sponsored by TEQIP-MHRD, Government of India..

e-Yantra (IIT-B) Pick & Place Robot

SLIET Longowal

with Team Endeavour under Dr. Amrik Singh

2018

Designed and built an autonomous robot for complex grid navigation and property-based object sorting; project sponsored by MHRD, Government of India.

UAV & Data Relay System:

SLIET Longowal

Semi-autonomous Unmanned Aerial Vehicle with Data Relaying Capabilities

2017

Scratch-built a quadcopter with a custom-programmed AtMega328p flight controller and MPU6050 IMU, achieving self-stabilization without GPS. Developed a long-range telemetry link using nRF24L01+ modules to transmit real-time flight data to a ground station. Attempts for localization and mapping for complete autonomy were unsuccessful at that stage.

Embedded Electronics Portfolio

SLIET Longowal

Various other robotics projects

2016-2020

Designed and implemented multiple robotics projects across Arduino, STM32, and RaspberryPi platforms, including autonomous maze solvers, obstacle avoidance systems, forward and inverse kinematics based robotic arm for assisting in soldering, sun-tracking solar arrays, and custom scale model aircraft.

Technical Skillset

Experimental Techniques: High-speed imaging, Particle Image Velocimetry (PIV), Particle Tracking Velocimetry (PTV), Planar Laser Induced Fluorescence (PLIF), Quantitative Smoke Visualization (QSV), Flow visualization using smoke/dye, shadowgraphy, Pitot tube measurements, Flow measurements with ultrasonic flow meters, Calibration and application of strain gauges and thermocouples, various machining processes and fabrication.

Programming and analysis: MATLAB, C++, Embedded C, L^AT_EX, python and basic web development.

Computational Fluid Dynamics: CFD Simulations using RANS and unsteady RANS, OpenFOAM (programming), Ansys Fluent (UDF).

Platforms/Softwares: OpenFOAM, Ansys Fluent, Ansys Mechanical, Fusion 360, CATIA, Arduino, Atmel/Microchip Studio.

Operating Systems: Linux and Windows.

Scholarships and Awards

Research Scholar Travel Award (RSTA) granted by **Indian Institute of Technology Delhi** for attending the International Congress of Theoretical and Applied Mechanics (**ICTAM**) 2024 in Daegu, South Korea.

Research Excellence Travel Award (RETA) granted by **Indian Institute of Technology Delhi** for attending the 78th annual meeting of the American Physical Society Division of Fluid Dynamics (**APS DFD**) 2025 in Houston, Texas, USA.

Awarded the APS DFD Travel Grant 2023 of \$1000 (could not avail due to visa issue)

Junior and Senior Research Fellowship by MHRD, Govt. of India.

Winner/Runner up of various robotics competitions during undergrad (including robotic arm, automation in agriculture, aerial robotics, multi-rotors, fixed wing model aircrafts etc.)

Publications

Cholemari, M. R., & **Vishnubhatla, S. H.** (2025). Analysis of an accurate iterative technique with attenuation compensation in planar laser induced fluorescence. *Measurement Science and Technology*, 36(1), 015301. [DOI](#).

Cholemari, M. R., **Vishnubhatla, S. H.**, & Krishnan, A. (2024). On the errors and their mitigation in planar laser induced fluorescence. *Measurement Science and Technology*, 35(7), 075206. [DOI](#).

Vishnubhatla, S. H., & Cholemari, M. R., Experimental Observations on Drag Reduction in a Horizontal Turbulent Pipe Flow Using Suspended Rigid Particles. (*Under Review*).

Vishnubhatla, S. H., & Cholemari, M. R., A Near-wall Projection Method to Measure Mean Wall Shear Stress using PIV in Standard Wall-bounded Flows (*Submitted*).

Vishnubhatla, S. H., & Cholemari, M. R., Drag Reduction in Particle Laden Turbulent Flow in a Vertical Pipe (*to be submitted*).

Vishnubhatla S. H. , OpenPTU: A Low-Cost, Open-Source Programmable Timing Unit with Sub-Microsecond Precision for PIV (*Submitted*).

Conferences

Vishnubhatla, S. H., & Cholemari, M. R., (2025). Intensity-Based Image Segmentation (IBIS) for Flow Diagnostics in Suspensions. *Bulletin of the American Physical Society*.

Vishnubhatla S. H., & Cholemari M. R., Drag Reduction using Particles as Shear Free Surfaces in a Gravity Driven Turbulent Pipe Flow, In *Proceedings of International Congress of Theoretical and Applied Mechanics (ICTAM 2024)*, Daegu, South Korea, pp. 1292-1293, ISBN:979-11-984760-9-8

Krishnan A., **Vishnubhatla, S. H.**, & Cholemari, M. R., (2023). Flux enhancement by Shear Free Surfaces in Turbulent Convection. *Bulletin of the American Physical Society*.

Vishnubhatla, S. H., & Cholemari, M. R., (2023). Drag Reduction using particles as Shear Free Surfaces in a turbulent pipe flow. *Bulletin of the American Physical Society*.

Vishnubhatla S. H., & Cholemari, M. R., Experimental Observations on Turbulent Drag Reduction using Shear Free Surfaces. In *Fluid Mechanics and Fluid Power, Volume 6. FMFP 2022*. Lecture Notes in Mechanical Engineering. Springer Nature Singapore (2022).

Beyond Research

My hobbies include long distance cycling, touring, robotics, roller skating, playing chess, playing badminton, song writing, composing and producing music and songs, singing, playing instruments like keyboard, guitar and flute.